

From Lexicons to LLMs: A Context-Aware Hybrid Pipeline for Fact vs. Opinion Classification

Mouad Asbayou¹

¹Graz University of Technology, Graz, Austria

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Abstract

This paper compares traditional Natural Language Processing (NLP) techniques and a Retrieval-Augmented Generation (RAG) framework for classifying text documents as facts or opinions. In Stage 1, we combined 6 linguistic markers with 300 TF-IDF features to train standard machine learning classifiers. A raw TF-IDF Logistic Regression established a baseline at 81.50% accuracy, which our Stage 1 Random Forest classifier matched using a PMI-calibrated Knowledge Base. While achieving the same 81.50% accuracy, the Knowledge Base approach reduced cross-validation variance and improved linguistic interpretability. In Stage 2, we integrated RAG, few-shot learning, and Chain-of-Thought (CoT) prompting using the `ministral-3:14b` model. To prevent class imbalances from vector space queries, we used a balanced retrieval strategy ($K = 4$, extracting 2 facts and 2 opinions) alongside a minimum similarity fallback threshold. This RAG configuration achieved a macro F_1 score of 94.14% and an accuracy of 94.15%. Our results show that while traditional NLP features are highly efficient for local environments, a fine-tuned RAG pipeline with Chain-of-Thought reasoning handles the subtle semantic shifts between facts and opinions much more accurately.

1 Introduction

Distinguishing between factual statements and subjective opinions is important for tasks like

misinformation detection, argument mining, and information retrieval. Automating this classification is difficult because identical words appear in both factual and opinionated contexts, and subjective signals often depend heavily on their local sentence structure.

This work presents a two-stage pipeline to classify text as facts or opinions. In Stage 1, we build a context-aware knowledge base (KB) using a manually curated lexicon, cancellation rules adapted from the VAGO expert system (Casanova et al., 2024), and statistical term weighting via Pointwise Mutual Information (PMI). We then extract linguistic features, part-of-speech (POS) ratios, and sentiment scores from this KB to train classical machine learning classifiers. In Stage 2, we implement a Large Language Model (LLM) within a Retrieval-Augmented Generation (RAG) framework (Lewis et al., 2020). This stage combines the Stage 1 knowledge base with few-shot prompting (Brown et al., 2020) and Chain-of-Thought (CoT) reasoning.

The dataset contains 855 crowdsourced sentences from web, news, and media texts. To focus our models on stylistic patterns rather than topical features, sentences were randomly shuffled before partitioning. While the training, testing, and validation splits share similar topics and vocabularies, sentence-level deduplication guarantees no text overlaps between the reference index and the evaluation sets. We evaluate both stages against clear baselines, prioritizing reproducibility and linguistic interpretability.

2 Related Work

Subjectivity analysis has been studied in NLP for over two decades. Riloff and Wiebe (2003) used a bootstrapping approach to learn linguistic patterns for subjective expressions and automatically identify subjective sentences. This work led to the formalization of annotation schemes for opinions by Wiebe et al. (2005), who released the MPQA Opinion Corpus. For broader opinion mining, Pang and Lee (2008) surveyed sentiment-aware classification techniques. A key finding relevant to our Stage 1 pipeline is that binary term presence outperforms term frequency for sentiment classification, which motivates our choice of binary TF-IDF features.

The introduction of pre-trained transformer models changed text classification. Devlin et al. (2019) showed that fine-tuning deep bidirectional representations achieves strong results across various NLP tasks. More recently, Brown et al. (2020) demonstrated that large generative models can perform competitive classification via few-shot prompting without requiring weight updates. Instruction-tuned architectures, such as the Llama and Mistral models, show strong performance in capturing subtle syntactic and semantic shifts (Jiang et al., 2023; Touvron et al., 2023). This capability is further improved when forcing the model to generate intermediate rationales before outputting a label, as formalized by Chain-of-Thought (CoT) prompting (Wei et al., 2022).

The closest benchmark to our task is the CLEF CheckThat! 2024 subjectivity detection track (Struß et al., 2024), which classifies whether a sentence is subjective or objective. The top system, HYBRINFOX (Casanova et al., 2024), combined a fine-tuned RoBERTa model with VAGO, an expert system that applies vagueness-cancellation rules to neutralize subjective signals based on local context. Our Stage 1 KB adapts this cancellation logic to a no-transformer setup. For Stage 2, our approach is motivated by Lewis et al. (2020), who showed that Retrieval-Augmented Generation (RAG) effectively improves LLM performance by providing external context.

3 Materials and Methods

3.1 Dataset

The dataset contains 855 sentences collected and labeled by course participants across topics like artificial intelligence, science, politics, and sports. Labels are divided into two binary classes: *Fact* (443 samples, 51.8%) and *Opinion* (412 samples, 48.2%). A separate, unlabeled validation set of 222 sentences was used for the final evaluation.

3.2 Stage 1: Knowledge Base Design

3.2.1 Lexicon Design

We built a lexicon covering six linguistic categories. Opinion-signaling categories include: *epistemic verbs* (*think, believe, seem*) to express uncertainty (Riloff and Wiebe, 2003); *evaluative adjectives* (*dangerous, misleading*) for value judgments (Wiebe et al., 2005); *hedging adverbs* (*probably, arguably*); *deontic modals* (*should, must*) for normative claims; and *intensifiers* (*extremely, utterly*). Fact-signaling categories include: *attribution verbs* (*found, reported, confirmed*) to indicate sources (Wiebe et al., 2005); *quantitative markers* (*percent, study, evidence*); and *citation phrases* (*according to*).

3.2.2 Cancellation Rules

Following the VAGO expert system framework (Casanova et al., 2024), we use three rules to neutralize subjective signals when terms appear in objective contexts:

- **Quotation cancellation:** Opinion words inside quotation marks are treated as factual reports of external statements rather than the author’s personal opinion.
- **Numeric cancellation:** Vague adjectives lose their subjective signal if they appear within a ± 30 character window of a number (e.g., *high* vs. *72% high*).
- **Attribution cancellation:** Opinion words are neutralized if they occur within 60 characters of an attribution trigger like *found that*.

A limitation is that these rules do not distinguish reporting quotes from scare quotes (e.g., “*experts*”), which would require dependency parsing outside our no-transformer pipeline.

3.2.3 PMI Calibration

Lexicon terms are weighted using Pointwise Mutual Information (PMI):

$$\text{PMI}(t, c) = \log \frac{P(t | c)}{P(t)} \quad (1)$$

where t is a term and $c \in \{\text{fact}, \text{opinion}\}$. We apply add- k smoothing ($k=0.1$) to handle missing values and exclude terms appearing fewer than 5 times to prevent noise.

3.3 Stage 1: Feature Extraction

We extract four distinct feature groups from each sentence:

KB features (9 dimensions): Raw and adjusted density scores for opinions and facts, the cancellation delta, first-person pronoun density, and question mark counts. The cancellation delta measures opinion words neutralized by objective contexts.

POS ratios (4 dimensions): Adjective, adverb, proper noun, and verb ratios calculated using NLTK. Adjective and adverb frequencies track descriptive language, while the proper noun ratio acts as a lightweight proxy for entity density. Named Entity Recognition (NER) was excluded because proper noun ratios provide equivalent information at a much lower computational cost.

Sentiment features (4 dimensions): Polarity and subjectivity scores from TextBlob, alongside compound and absolute compound scores from VADER (Hutto and Gilbert, 2014; Pang and Lee, 2008).

TF-IDF (300 dimensions): Binary unigram and bigram features ($\text{min_df}=3$). We use binary presence weights rather than raw term frequencies based on findings by Pang and Lee (2008).

To prevent data leakage, the TF-IDF vocabulary matrix and scaling transforms are fit strictly on the cross-validation training folds and applied downstream to the test partitions.

3.4 Stage 1: Classifiers and Evaluation

We evaluate three classifiers: Logistic Regression (LR), Linear SVM, and a Random Forest (RF) with 200 trees. All dense features are z-score standardized before training. Performance is verified using 5-fold stratified cross-validation with macro F1 as the primary metric.

Baseline 1 is a majority-class predictor that always outputs *Fact*. **Baseline 2** uses raw TF-IDF features with a Logistic Regression classifier, skipping the handcrafted KB features. **Model 1** integrates binary KB weights, while **Model 2** implements the PMI-calibrated weights to isolate the specific impact of statistical term weighting.

3.5 Stage 2: LLM-Based Classification

3.5.1 Approach Overview

Stage 2 classifies sentences using the `ministral-3:14b` model via an OpenAI-compatible API. The model is evaluated in a Retrieval-Augmented Generation (RAG) framework (Lewis et al., 2020) combined with few-shot prompting (Brown et al., 2020) and Chain-of-Thought (CoT) reasoning. A zero-shot prompt using the same model serves as the baseline to isolate the impact of retrieval context.

3.5.2 Semantic Retrieval

We retrieve similar labeled examples from the training data using 384-dimensional dense vectors generated by the `all-MiniLM-L6-v2` sentence transformer (Reimers and Gurevych, 2019). To prevent the retriever from biasing the prompt toward the majority class, we use a **balanced retrieval** strategy that extracts the top- $K/2$ facts and top- $K/2$ opinions separately ($K=4$). A **similarity threshold** ($\tau=0.45$) acts as a fallback: if the best retrieved example falls below τ , retrieval context is omitted, and the model defaults to zero-shot classification using only the Stage 1 KB signals.

3.5.3 Prompt Design

Prompts concatenate three components: (1) $K=4$ balanced few-shot examples; (2) the structured linguistic signals from Stage 1 (density scores, cancellation values, and pronoun counts); and (3) a CoT instruction forcing step-by-step reasoning. The prompt explicitly instructs the model to provide its final decision as a single word on the last line. Complete templates are documented in Appendix B.

3.5.4 Label Parsing

Because target terms like *fact* or *opinion* regularly appear within the model’s reasoning steps, simple keyword searches introduce errors. We use a three-priority extraction parser: (1) check for explicit conclusion markers (e.g., *Answer*); (2) scan lines from the bottom for a standalone label; and (3) fall back to the absolute last occurrence of a target word.

3.5.5 Evaluation Setup

Stage 2 models are tested on the exact same 20% stratified held-out split ($n=171$) used in Stage 1. The RAG retrieval index is built exclusively from the remaining 80% training portion ($n=684$) to prevent evaluation leakage.

4 Results

4.1 Stage 1

Table 1 reports 5-fold cross-validated accuracy and macro F1 for all Stage 1 models. Table 2 reports the ablation study isolating each KB component’s contribution.

Table 1: Stage 1 results (5-fold CV, mean $\pm$ std).

Model	Accuracy	F1 macro
Baseline 1 (majority class)	0.518 $\pm$ 0.003	0.341 $\pm$ 0.001
Baseline 2 (TF-IDF + LR)	0.815 $\pm$ 0.027	0.815 $\pm$ 0.027
Model 1 – Binary KB + LR	0.808 $\pm$ 0.011	0.808 $\pm$ 0.011
Model 1 – Binary KB + SVM	0.800 $\pm$ 0.015	0.800 $\pm$ 0.016
Model 1 – Binary KB + RF	0.801 $\pm$ 0.021	0.800 $\pm$ 0.021
Model 2 – PMI KB + LR	0.814 $\pm$ 0.018	0.813 $\pm$ 0.018
Model 2 – PMI KB + SVM	0.797 $\pm$ 0.009	0.796 $\pm$ 0.010
Model 2 – PMI KB + RF	0.815 $\pm$ 0.014	0.815 $\pm$ 0.014

Table 2: Ablation study: incremental contribution of each KB component (LR, 5-fold CV).

Variant	F1 macro
A – POS + Sentiment + TF-IDF (no KB)	0.7985
B – + Binary KB, no cancellation	0.8063
C – + Binary KB, with cancellation	0.8075
D – + PMI KB, with cancellation	0.8133
Cancellation contribution (C – B)	+0.0012
PMI calibration contribution (D – C)	+0.0058
Total KB contribution (D – A)	+0.0148

4.2 Stage 2

Table 3 compares the performance of the zero-shot baseline against the finalized RAG pipeline on the held-out 20% split ($n=171$).

Table 3: Stage 2 results (held-out 20%, $n=171$).

Model	Accuracy	F1 macro
Zero-shot baseline	0.924	0.923
RAG + Few-Shot + CoT	0.942	0.941

5 Discussion

5.1 Stage 1

The strong performance of Baseline 2 ($F_1 = 0.815$) relative to the full models (best $F_1 = 0.815$) in Table 1 reveals that raw lexical patterns captured by TF-IDF account for most of the discriminative signal in this dataset. Notably, adding binary KB features without PMI calibration (Model 1) slightly underperforms Baseline 2, suggesting that equally-weighted KB features introduce noise that TF-IDF alone avoids. PMI calibration (Model 2) corrects this, recovering Baseline 2 performance while revealing lower cross-validation variability (± 0.014 versus ± 0.027).

The ablation study in Table 2 quantifies each component’s contribution. Cancellation rules contribute +0.0012 F_1 , yet modest but positive, confirming that context-aware neutralization of opinion signals adds some value. The gain is small because the crowdsourced dataset contains relatively very few sentences with the ambiguous reporting-quote or numeric-context

patterns that cancellation addresses; a news-heavy corpus such as CLEF CheckThat! would likely amplify this contribution. PMI calibration contributes a larger $+0.0058 F_1$, confirming that data-driven term weighting is the most impactful KB component.

A notable misclassification example: *“Helium is an important driver of...”* is predicted Opinion, while true label Fact. The phrase *important driver* triggers the evaluative adjective signal despite the purely descriptive usage, which illustrates a core limitation of token-level matching even with cancellation rules.

5.2 Stage 2

As shown in Table 3, the zero-shot baseline achieves an F_1 score of 0.923, showing that the underlying `ministral-3:14b` model accurately distinguishes facts from opinions without external context. Augmenting the model with our RAG and Chain-of-Thought pipeline improves performance to a macro F_1 score of 94.14% and a classification accuracy of 94.15%.

The evaluation metrics indicate that the balanced semantic retrieval strategy ($K = 4$, split evenly between facts and opinions) functions as intended. Qualitative inspection of retrieved examples suggests that the embedding model frequently retrieves sentences sharing similar structural patterns, such as citation-based statements or numerical evidence rather than overlapping keywords. This allows the few-shot examples to serve as stable structural templates for the model’s in-context learning.

During pipeline optimization, a bottleneck was identified regarding the token generation limit. In early validation runs, the prompt required step-by-step reasoning but generation was initially restricted to `max_tokens=200`. Qualitative log inspection showed that the model routinely hit this token limit mid-sentence while explaining its reasoning, cutting off before outputting the actual classification label. This truncation forced the parsing script to fall back to the last mentioned keyword within the incomplete text fragment, creating artificial classification errors. Increasing the generation ceiling to `max_tokens=512` and updating the prompt to encourage short and concise

reasoning points resolved this issue, allowing the pipeline to reach its final accuracy.

6 Conclusion

This work presented a two-stage pipeline for classifying text as facts or opinions. In Stage 1, we built a context-aware knowledge base (KB) combining a linguistic lexicon with VAGO-inspired cancellation rules and PMI calibration. While the final Stage 1 model achieved similar performance to the TF-IDF baseline, the ablation study shows that the knowledge base features contributed measurable improvements when evaluated independently ($+0.0148 F_1$). These features also improve linguistic interpretability by exposing which signals influence the classification decision.

In Stage 2, the `ministral-3:14b` model achieved a zero-shot baseline F_1 score of 0.923, demonstrating strong out-of-the-box capabilities for subjectivity detection. Combining this model with a balanced semantic retrieval RAG pipeline, Stage 1 linguistic signals, and Chain-of-Thought reasoning pushed the final performance to a macro F_1 score of 94.14% once token limit bottlenecks and parsing scripts were optimized.

A key finding is that lexical information alone captures much of the discriminative signal in this dataset, as demonstrated by the strong TF-IDF baseline. The proposed knowledge base contributes interpretability and modest performance gains through context-aware linguistic features, while the LLM-based RAG pipeline is better able to handle subtle semantic distinctions that remain challenging for feature-engineered approaches.

The comparison highlights a clear engineering trade-off: Stage 1 is computationally efficient, localized, and highly interpretable when labeled data is present, whereas Stage 2 requires no task-specific training but depends heavily on prompt engineering and infrastructure tuning. Future work should evaluate the cancellation-rule KB on broader news benchmarks, such as the CLEF CheckThat! 2024 dataset, where complex document structures would likely amplify the impact of context-aware cancellation rules.

Limitations

The pipeline operates exclusively on English text; extending to other languages would require rebuilding the lexicon with culturally appropriate signal terms.

The PMI weights are calibrated on a small, crowdsourced dataset (855 sentences), which may introduce noise and limit generalizability to other domains.

The quotation cancellation rule cannot distinguish reporting quotes from scare quotes, occasionally suppressing genuine opinion signals.

Since the dataset is crowdsourced from a single course cohort, train and evaluation splits may share topical overlap, potentially inflating cross-validation performance relative to a truly out-of-distribution evaluation.

The relatively small improvement over the zero-shot baseline suggests that much of the classification capability originates from the underlying language model itself. Future work should evaluate the proposed retrieval and knowledge-base components on more challenging datasets where the baseline model performance is lower.

Furthermore, a distinct operational constraint within the Stage 2 framework relates to the strict generation allocation budget on public or shared API infrastructures. Complex prompting techniques (e.g., Chain-of-Thought) demand significant sequence space to build logical steps. If generation ceilings are constrained to short lengths, the text string risks getting cut off mid-sentence, preventing terminal classification extraction and causing downstream parsers to default to unreliable text parsing rules.

Ethical Considerations

Automated fact versus opinion classification carries inherent risks of misuse if deployed as an automated censorship or content moderation tool, particularly within political discourse lacking human oversight. Structurally, the pipeline is trained on a crowdsourced dataset that reflects the subjective biases and labeling heuristics of its contributors. Furthermore, the Stage 2 architecture relies on an externally

hosted large language model, which incurs an environmental cost regarding energy consumption during inference. Consequently, all classification outputs from this system should be treated strictly as downstream decision support rather than absolute ground truth.

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A Implementation Details

Stage 1 scripts: `knowledge_base.py` (lexicon and KnowledgeBase class), `fit_kb.py` (PMI weight computation), `features.py` (feature extraction), `baseline1.py`, `baseline2.py`, `model1.py`, `model2.py`, `ablation.py`, `predict_val.py`.

Stage 2 scripts: `embedder.py` (sentence transformer embedding and caching), `retriever.py` (cosine similarity retrieval), `s2_baseline.py` (zero-shot baseline), `s2_rag.py` (full RAG pipeline), `s2_predict_val.py` (validation set predictions).

B Prompts

Stage 2 System Prompt

You are an expert linguist specializing in distinguishing factual statements from opinions.

A FACT is a statement that:

- Can be verified against external sources
- References specific entities, numbers, dates, or events
- Uses neutral, objective language
- Attributes claims to named sources

An OPINION is a statement that:

- Expresses a personal belief, judgment, or evaluation
- Uses evaluative or emotionally charged language
- Makes normative claims about what ought to happen
- Cannot be objectively verified

Important: evaluative words inside quotation marks or next to numbers may not signal opinion.

Think step by step. Write your
final answer
on the very last line as exactly
one word:
Fact or Opinion.

Stage 2 User Prompt Structure

Similar labeled sentences (2
Facts, 2 Opinions):
[FACT]: "..."
[OPINION]: "..."

Linguistic signals detected:
- Opinion signals detected (
score: X.XX)
- Fact signals detected (score:
X.XX)
- [cancellation/first-person
notes if present]

Sentence to classify:
"[input sentence]"

Think step by step, then write
your final
answer on the last line (Fact or
Opinion).